

Started on Wednesday, 14 January 2026, 10:30 AM

State Finished

Completed on Wednesday, 14 January 2026, 11:05 AM

Time taken 35 mins 15 secs

Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Q1-Let A and B be 3×3 matrices. Suppose that

$$(A^T - 3I)^{-1} = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \quad \text{and } |2B| = 8$$

where I is the 3×3 identity matrix and $|\cdot|$ denotes the determinant.

Answer the following **multiple-choice** items:

(1) Find the matrix A :

- ☐ A. $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
- ☒ B. $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ ✓
- ☐ C. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$
- ☐ D. $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Your answer is correct.

The correct answer is: B) $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Question 2

Correct

Mark 1.00 out of 1.00

Q1-Let A and B be 3×3 matrices. Suppose that

$$(A^T - 3I)^{-1} = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \quad \text{and } |2B| = 8$$

(2) Find $\det(B^{-1})$:

Select one:

- ☐ A. 8
- ☐ B. 2
- ☒ C. 1 ✓
- ☐ D. 1/8

Your answer is correct.

The correct answer is: 1

Question 3

Correct

Mark 1.00 out of 1.00

Q1-Let A and B be 3×3 matrices. Suppose that

$$(A^T - 3I)^{-1} = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \quad \text{and } |2B| = 8$$

Find $|2AB|$:

Select one:

- ☒ A. 1920 ✓
- ☐ B. 1915
- ☐ C. 1905
- ☐ D. 1900

Your answer is correct.

The correct answer is: 1920

Question 4

Correct

Mark 2.00 out of 2.00

Q2 Use the given table to decode the message,
20 , 1 , 27 , 5 , 8 , 8 , 22 , 13 , 14 , 14 , 38 , 23

using the matrix , $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

_ = 0	I = 9	R = 18
A = 1	J = 10	S = 19
B = 2	K = 11	T = 20
C = 3	L = 12	U = 21
D = 4	M = 13	V = 22
E = 5	N = 14	W = 23
F = 6	O = 15	X = 24
G = 7	P = 16	Y = 25
H = 8	Q = 17	Z = 26

What is the decoded message?

Select one:

- ☐ A. **SAVE HIM NEW**
- ☐ B. **SAVE HER NOW**
- ☒ C. **SAVE HIM NOW** ✓
- ☐ D. **SAVE HIM NOT**

Your answer is correct.

The correct answer is: **SAVE HIM NOW**

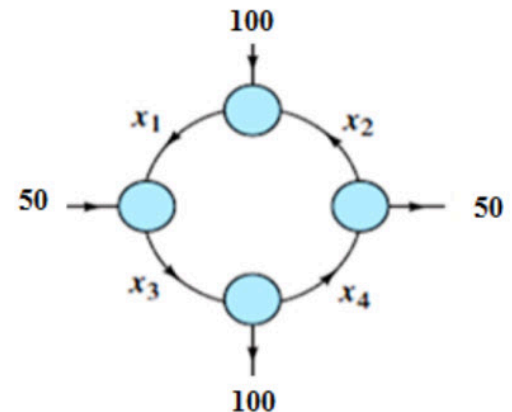
Question 5

Correct

Mark 1.00 out of 1.00

Q3

During the design of a new square in Egypt's New Administrative Capital, the smart-city team is testing an AI-based traffic digital twin. The square is a circular one-way loop with four junctions (North, East, South, West). The AI system labels the internal one-way flows on the loop as x_1, x_2, x_3, x_4 (vehicles/hour), moving clockwise between adjacent junctions.



External measured flows (from smart cameras + sensors) are:

- North junction: 100 vehicles/hour enter the loop from a main avenue.
- South junction: 100 vehicles/hour exit the loop to a main avenue.
- West junction: 50 vehicles/hour enter the loop from a feeder road.
- East junction: 50 vehicles/hour exit the loop to a feeder road.

Assume steady-state flow and conservation at each junction (Input = Output).

(A) Which linear system correctly models the loop using flow conservation?

- ☐ A.
$$\begin{cases} 100 + x_1 = x_2 \\ 50 + x_3 = x_1 \\ x_4 = 100 + x_3 \\ x_2 = 50 + x_4 \end{cases}$$
- ☐ B.
$$\begin{cases} 100 + x_2 = x_1 \\ 50 + x_1 = x_3 \\ x_3 = 50 + x_4 \\ x_4 = 100 + x_2 \end{cases}$$
- ☒ C.
$$\begin{cases} 100 + x_2 = x_1 \\ 50 + x_1 = x_3 \\ x_3 = 100 + x_4 \\ x_4 = 50 + x_2 \end{cases}$$
 ✓
- ☐ D.
$$\begin{cases} 100 + x_2 = x_3 \\ 50 + x_1 = x_2 \\ x_3 = 100 + x_4 \\ x_4 = 50 + x_1 \end{cases}$$

Your answer is correct.

The correct answer is:
$$\begin{cases} 100 + x_2 = x_1 \\ 50 + x_1 = x_3 \\ x_3 = 100 + x_4 \\ x_4 = 50 + x_2 \end{cases}$$

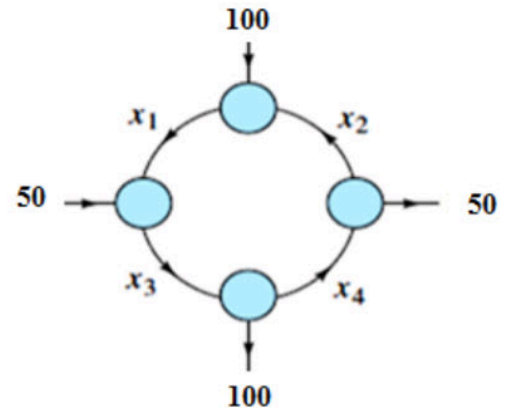
Question 6

Correct

Mark 1.00 out of 1.00

Q3

During the design of a new square in Egypt's New Administrative Capital, the smart-city team is testing an AI-based traffic digital twin. The square is a circular one-way loop with four junctions (North, East, South, West). The AI system labels the internal one-way flows on the loop as x_1, x_2, x_3, x_4 (vehicles/hour), moving clockwise between adjacent junctions.



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- West junction: 50 vehicles/hour enter the loop from a feeder road.
- East junction: 50 vehicles/hour exit the loop to a feeder road.

Assume steady-state flow and conservation at each junction (Input = Output).

(B) From an AI/data-science viewpoint, what does it mean if the system has infinitely many solutions?

- ☒ A. The measurements constrain the flows, but one internal loop flow is still a free parameter (multiple valid states fit the same boundary data). ✓
- ☐ B. The sensors are wrong and must be replaced.
- ☐ C. The loop must be closed permanently.
- ☐ D. The network is unstable and cannot be controlled.

Your answer is correct.

The correct answer is: **The measurements constrain the flows, but one internal loop flow is still a free parameter (multiple valid states fit the same boundary data).**

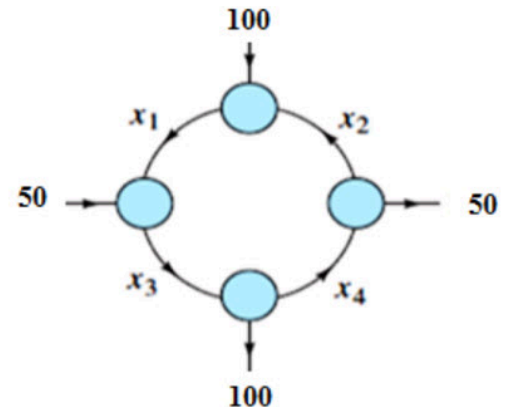
Question 7

Correct

Mark 1.50 out of 1.50

Q3

During the design of a new square in Egypt's New Administrative Capital, the smart-city team is testing an AI-based traffic digital twin. The square is a circular one-way loop with four junctions (North, East, South, West). The AI system labels the internal one-way flows on the loop as x_1, x_2, x_3, x_4 (vehicles/hour), moving clockwise between adjacent junctions.



External measured flows (from smart cameras + sensors) are:

- North junction: 100 vehicles/hour enter the loop from a main avenue.
- South junction: 100 vehicles/hour exit the loop to a main avenue.
- West junction: 50 vehicles/hour enter the loop from a feeder road.
- East junction: 50 vehicles/hour exit the loop to a feeder road.

Assume steady-state flow and conservation at each junction (Input = Output).

(C) If the AI controller sets $x_4 = 50$ vehicles/hour (e.g., by adaptive signals), what are the other flows?

- ☐ A. $(x_1, x_2, x_3, x_4) = (50, 100, 0, 150)$
- ☐ B. $(x_1, x_2, x_3, x_4) = (150, 0, 100, 50)$
- ☒ C. $(x_1, x_2, x_3, x_4) = (100, 0, 150, 50)$ ✓
- ☐ D. $(x_1, x_2, x_3, x_4) = (100, 50, 150, 0)$

Your answer is correct.

The correct answer is: $(x_1, x_2, x_3, x_4) = (100, 0, 150, 50)$

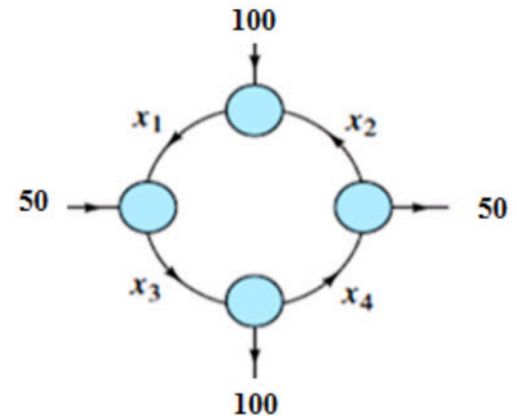
Question 8

Correct

Mark 1.50 out of 1.50

Q3

During the design of a new square in Egypt's New Administrative Capital, the smart-city team is testing an AI-based traffic digital twin. The square is a circular one-way loop with four junctions (North, East, South, West). The AI system labels the internal one-way flows on the loop as x_1, x_2, x_3, x_4 (vehicles/hour), moving clockwise between adjacent junctions.



External measured flows (from smart cameras + sensors) are:

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- West junction: 50 vehicles/hour enter the loop from a feeder road.
- East junction: 50 vehicles/hour exit the loop to a feeder road.

Assume steady-state flow and conservation at each junction (Input = Output).

(D) In the digital twin, the AI learns the free parameter r such that $x_4 = r$. Which parametric solution matches the model?

- ☐ A. $x_1 = r, x_2 = r, x_3 = r, x_4 = r$
- ☐ B. $x_1 = 50 - r, x_2 = 50 + r, x_3 = 100 - r, x_4 = r$
- ☒ C. $x_1 = r + 50, x_2 = r - 50, x_3 = r + 100, x_4 = r$ ✓
- ☐ D. $x_1 = r - 50, x_2 = r + 50, x_3 = r - 100, x_4 = r$

Your answer is correct.

The correct answer is: $x_1 = r + 50, x_2 = r - 50, x_3 = r + 100, x_4 = r$